Final paper

Are Catastrophe Bonds Really Zero-Beta Assets?

Diversification Benefits in Normal Markets and Financial Crises

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1. Introduction

A catastrophe (CAT) bond lets an insurer, a reinsurer or a government pass part of its exposure to a natural disaster to capital-market investors. The sponsor pays a premium, the investors put up principal that is held as collateral, and if a specified hurricane, earthquake or other event occurs before maturity, part or all of that principal goes to the sponsor (Cummins, 2008). Froot (2001) showed that catastrophe reinsurance in the 1990s was priced far above expected losses and traced the gap to frictions in the supply of reinsurance capital. Securitization was one answer: it opens catastrophe risk to investors whose combined wealth is much larger than the capital of the reinsurance industry.

To attract those investors, the market offered a simple argument. Whether a hurricane hits Florida or an earthquake hits Tokyo does not depend on the stock market, so the losses on a CAT bond should be unrelated to the losses on stocks and corporate bonds. In CAPM terms, CAT bonds would be zero-beta assets, and their spread over the risk-free rate would pay investors for a risk that adds nothing to the systematic risk of a diversified portfolio.

The argument concerns the natural event, while the investor holds a financial security. A CAT bond is a contract with a special purpose vehicle, its principal sits in a collateral account, its price comes from dealer quotes, and it is held by funds that may need cash at the same moment as everyone else. Each of these features can carry financial stress into the bond when no catastrophe has happened. Whether they do so is an empirical question, and the published answers disagree.

Carayannopoulos and Perez (2015) find that CAT bonds behaved as zero-beta assets outside crisis periods but became correlated with the market after the collapse of Lehman Brothers. Drobetz, Schröder and Tegtmeier (2020) find that CAT bonds are an effective diversifier but a poor hedge, and that they acted as a strong safe haven against extreme stock market declines only after the financial crisis. Haffar and Le Fur (2022) find that, of the assets they study, CAT bonds have the highest risk-adjusted performance and add the least to the probability of extreme joint movements with an equity portfolio. Other studies push in both directions. Gürtler, Hibbeln and Winkelvos (2016) conclude from the link between CAT bond premiums and corporate bond spreads that CAT bonds should not be regarded as zero-beta securities, while Feng and Zeng (2025) find that CAT bonds acted as a safe haven against stocks after the COVID-19 outbreak.

The difference matters for a pension fund or an insurer deciding how much to invest. A diversifier lowers the variance of a portfolio on average. A safe haven holds up when the rest of the portfolio is in a crash. An investor who buys CAT bonds for the second property when they only have the first is exposed at the worst possible moment.

My research question is: **to what extent do CAT bonds retain their diversification benefits during financial stress, and why do published studies reach different answers?** I argue that most of the disagreement has little to do with catastrophe risk. It comes from four sources: the property being tested (diversifier, hedge or safe haven), the collateral structure in force during the sample, the type of stress (a shortage of funding or a fall in equity prices), and the way CAT bond

returns are measured. As my personal contribution, I decompose the CAT bond return into an insurance core, which is independent of markets, and a financial wrapper, whose market exposure changes with the bond's structure and with the stress episode. This decomposition sorts the episodes studied in the literature into four regimes, predicts the result each study should find in each regime, and leads to proposals for investors and for the design of future tests.

Section 2 explains how CAT bonds work, defines the three properties that the literature tests, presents the three articles in detail and adds the studies that diverge from them. Section 3 contains my analysis: the decomposition, the four regimes, the measurement problems and the proposed solutions. Section 4 concludes.

2. Literature review

2.1 How a CAT bond works, and where financial risk can enter

A sponsor sets up a special purpose vehicle (SPV) that sells notes to investors and signs a reinsurance contract with the sponsor. The proceeds of the notes are placed in a collateral trust account. Investors receive the return earned by the collateral plus a fixed risk spread paid out of the sponsor's premium. If a qualifying event occurs, the collateral is released to the sponsor to pay its claims and investors lose part or all of their principal (Cummins, 2008). The trigger can be the sponsor's own losses (indemnity), an industry loss index, a modelled loss, or a physical parameter such as wind speed or ground motion. Indemnity triggers protect the sponsor best but expose investors to the sponsor's underwriting and claims practices. Index and parametric triggers reduce that moral hazard at the cost of basis risk for the sponsor (Cummins, 2008; Cummins and Weiss, 2009).

Several parts of this structure have nothing to do with nature. One is the **collateral**. Before 2009, many deals did not simply hold government securities. The collateral was invested in other assets, and an investment bank promised the SPV a money-market return through a total return swap, making up any shortfall in the value of the collateral. Lehman Brothers Special Financing was the swap counterparty for four CAT bonds: Ajax Re, Carillon, Newton Re and Willow Re. When Lehman failed in September 2008, the collateral behind these bonds lost its guarantee. Ajax Re paid all its interest but could not repay its principal in full at maturity in May 2009 because the collateral assets were worth less than the notes (Artemis, 2009). Carayannopoulos and Perez (2015) attribute the crisis effect on CAT bonds mainly to weaknesses in the composition and structure of the trust account, and they observe that new bonds issued from 2009 onward used improved structures. In practice, post-crisis deals hold short-term Treasury instruments in trust, without a bank guaranteeing their value.

Another is **liquidity**. CAT bonds trade over the counter and not very often. Using 9,393 secondary-market trades reported to TRACE between 2015 and 2019, Herrmann and Hibbeln (2023) find that liquidity is lower for riskier bonds and for bonds far from maturity, and they estimate that the liquidity premium accounts for about a fifth of the observed CAT bond yield

spread. A premium of that size can move when dealers and investors run short of capital, which is also when stock prices fall (Brunnermeier and Pedersen, 2009).

Measurement is a further source of financial noise. Most empirical studies use Swiss Re's CAT bond indices. Drobetz et al. (2020) describe the index they use as tracking secondary-market bid indications that Swiss Re provides to investors each week. An index built from dealer quotes tends to be smoothed: it absorbs news with a delay, which lowers its measured volatility and its contemporaneous correlation with other markets (Getmansky, Lo and Makarov, 2004).

2.2 Diversifier, hedge and safe haven: what each test measures

The articles reviewed here use, explicitly or not, the definitions of Baur and Lucey (2010). A **hedge** is an asset that is uncorrelated or negatively correlated with another asset or portfolio on average. A **diversifier** is positively, but not perfectly, correlated with it on average. A **safe haven** is uncorrelated or negatively correlated with it during extreme market declines. Baur and McDermott (2010) distinguish strong and weak versions: a strong hedge or safe haven is negatively correlated with the market, a weak one is uncorrelated.

These properties are distinct: an asset can be a good diversifier and still fail as a safe haven. The zero-beta claim, which says the beta is zero at all times, is closer to "weak hedge plus weak safe haven" than to diversification, so evidence of diversification does not confirm it. The definitions are also about correlation, not about price. A statistical safe haven can lose money in a crash as long as its losses do not move with the market's. An investor who has to sell during a crisis cares about the price received, so the economic meaning of a safe haven is stricter than the statistical one.

The tools differ as well. A beta is the covariance with the market divided by the market's variance. A correlation scales the same covariance by both volatilities. A copula separates the dependence between two series from their individual distributions and can measure tail dependence, the probability that one series has an extreme loss given that the other has one. These tools can give different verdicts on the same data. Forbes and Rigobon (2002) showed one reason: when market volatility rises, the measured correlation rises even if the underlying relation between the two assets has not changed, so correlations estimated in crisis windows are biased upward unless they are corrected.

2.3 Carayannopoulos and Perez (2015): zero beta, except in a crisis

The first article asks two questions: are CAT bonds zero-beta investments, and do they diversify a portfolio? The authors study the joint dynamics of CAT bond returns and the returns of the stock, corporate bond and government bond markets with a multivariate GARCH model. From the estimated conditional covariances they compute dynamic hedge ratios, the conditional covariance with each market divided by that market's conditional variance. Because a hedge ratio defined this way is a beta, they read the series as time-varying betas.

The results depend on the period. Outside the crisis, the betas are indistinguishable from zero and CAT bonds look like the zero-beta asset of the textbook argument. With the collapse of Lehman Brothers, CAT bond returns became significantly correlated with the market. The authors add two qualifications. The crisis affected CAT bonds much less than it affected the other asset classes, so CAT bonds still diversified a portfolio. And CAT bond betas returned to their pre-crisis levels once bonds with improved structures were issued from 2009 onward, which the authors read as the market's approval of those structures.

The strength of the article is that it treats beta as a variable. The time-varying hedge ratios show when the zero-beta property broke down, and the timing points to the trust-account structure, since catastrophe risk itself did not change when Lehman failed. Its limit, for my question, is that a beta estimated on returns cannot say which part of the bond moved with the market. A fall in the value of Lehman-guaranteed collateral, a rise in the premium demanded by investors who needed cash, and a stale price catching up after a quiet period would all appear as a positive beta. The article also studies a single crisis, and an unusual one for CAT bonds, since a swap counterparty failed during it. Timing adds a further difficulty. Hurricane Ike made landfall in Texas on 13 September 2008, two days before Lehman Brothers filed for bankruptcy, so in weekly data a catastrophe loss and a financial shock can fall in the same observation. In a short crisis window such a coincidence carries a large weight, and a crisis beta is only interpretable once the two are separated.

2.4 Drobetz, Schröder and Tegtmeier (2020): a diversifier, and a safe haven only after the crisis

The second article widens the comparison to an international multi-asset portfolio. The authors pair the Swiss Re Global Unhedged CAT Bond Performance Index with six asset classes: global stocks, bonds, real estate, commodities, private equity and infrastructure, using weekly returns. Their method has two stages, following Bouri et al. (2017). First, they estimate dynamic conditional correlations (DCC) between the CAT bond index and each asset class. Second, they regress these correlations on dummy variables that mark the weeks in which the other asset's return falls in its extreme lower tail. The constant of the regression measures the average correlation, which settles the hedge-or-diversifier question. The coefficients on the dummies measure how the correlation changes in extreme weeks, which settles the safe-haven question.

According to the article, CAT bonds are a poor hedge, since their average correlation with the other asset classes is positive. They are an effective diversifier against all six classes. They are a strong safe haven against extreme declines of the stock market, but only during the post-crisis period. In the earlier sub-periods this property is absent.

This article is the closest to my question because it separates the three properties and tests them by sub-period. It also sharpens the puzzle. If the zero-beta argument were complete, the safe-haven property would hold in every period, because the probability of a hurricane does not depend on the state of financial markets. Two features of the design matter for the interpretation. The index is built from bid indications, not trades, so weekly correlations may understate co-

movement when quotes adjust slowly. And the extreme-week dummies identify stress through the other asset's return only. A week in which stock prices fall while funding markets work normally and a week of forced selling in a funding crisis are treated as the same event.

2.5 Haffar and Le Fur (2022): little extreme co-movement

The third article studies the dependence structure itself. The authors model each return series with a GARCH process and join the series with copulas, which lets the dependence change over time and differ between ordinary and extreme outcomes. Their portfolio combines CAT bonds with six stock markets: the CAC 40, the Dow Jones Industrial Average, the EURO STOXX 50, the FTSE 100, the Hang Seng and the Nikkei 225. The authors present the paper as the first to analyse the dependence structure and diversification value of CAT bonds with copulas.

Their findings are the most favourable of the three. CAT bonds display the highest risk-adjusted performance among the assets studied, they are one of the best diversifiers, and they are the asset that raises the probability of extreme co-movements with the benchmark portfolio the least.

The copula approach answers a question that correlation-based studies cannot answer directly: do the two assets crash together? On that question, the article supports the zero-beta view, and because the GARCH margins absorb changes in volatility, it is less exposed to the bias identified by Forbes and Rigobon (2002). The result has limits. The alternatives against which CAT bonds are ranked are stock indices, which depend strongly on each other and on an equity benchmark by construction, so ranking first among them on extreme co-movement is a modest test. The headline results describe the dependence structure as a whole. They do not tell an investor whether the probability of a joint extreme rises in a funding crisis like that of 2008. And if the CAT bond series is an index of quoted prices, as in most of this literature, its volatility is understated and its risk-adjusted performance overstated (Getmansky et al., 2004).

2.6 Other studies: divergent and supporting evidence

Two studies of CAT bond prices take the opposite side. Gürtler et al. (2016) analyse premiums in the secondary market and find that both the financial crisis and Hurricane Katrina had a significant effect on them. They also find a positive relation between corporate bond spreads and CAT bond premiums, a dependence that grew stronger during the crisis, and conclude that CAT bonds should not be regarded as zero-beta securities. Braun (2016), who models spreads at issuance, finds that apart from the expected loss, the covered territory, the sponsor and the reinsurance cycle, the spreads of comparably rated corporate bonds help explain CAT bond spreads. These two studies concern the price of risk and say nothing directly about realized losses. They show that the premium investors demand for catastrophe risk moves with the premium they demand for credit risk.

A recent study supports the favourable view for a later crisis. Feng and Zeng (2025) apply a DCC-GARCH model to an international multi-asset portfolio before and during the COVID-19 pandemic. They find that CAT bonds were a diversifier over the whole sample and a safe haven against stocks and private equity after the outbreak. The results hold with desmoothed returns,

which addresses the quote-smoothing problem described above, but the authors also find that hedge ratios are sensitive to extreme weather events. Market reports from March 2020 show what the stress looked like inside the CAT bond market: the Swiss Re price return index fell by 0.81 percent in the week to 20 March, with markdowns linked to bulk lists of bonds put up for sale (Insurance Insider ILS, 2020).

The same period produced a case in which the zero-beta argument fails by design. The World Bank's pandemic bonds, issued in 2017 for its Pandemic Emergency Financing Facility, put 16.67 percent of the principal of the Class A notes and all of the principal of the Class B notes at risk if a qualifying pandemic occurred (Government Actuary's Department, 2021). COVID-19 met the trigger conditions in 2020, during the same shock that hit equity markets. When the insured peril is itself a macroeconomic shock, the insurance losses and the market move together, and no collateral rule can separate them.

Table 1 compares the three main articles.

Table 1. The three articles compared

	Carayannopoulos and Perez (2015)	**Drobetz, Schröder and Tegtmeier (2020)**	**Haffar and Le Fur (2022)**
Question	Are CAT bonds zero-beta, and did the subprime crisis change this?	Are CAT bonds a diversifier, a hedge or a safe haven?	How do CAT bonds depend on equity markets, including in the tails?
Comparison assets	Stocks, corporate bonds, government bonds	Global stocks, bonds, real estate, commodities, private equity, infrastructure	Six stock indices (CAC 40, DJIA, EURO STOXX 50, FTSE 100, Hang Seng, Nikkei 225)
Method	Multivariate GARCH; dynamic hedge ratios read as betas	DCC, then regression of the DCC on extreme-week dummies	Copula-GARCH; time-varying and tail dependence
What counts as stress	The crisis period after Lehman's failure	Weeks of extreme decline in the other asset, by sub-period	Joint extreme outcomes in the tails
Statistic	Beta	Conditional correlation	Copula dependence
Main conclusion	Zero-beta only outside crises; less affected than other assets; betas back to pre-crisis levels after the 2009 structures	Poor hedge; effective diversifier; strong safe haven against stocks only post-crisis	Highest risk-adjusted performance; one of the best diversifiers; least increase in extreme co-movement

3. Analysis

3.1 What the articles actually disagree about

All the studies find that CAT bonds diversify a portfolio on average. Carayannopoulos and Perez show it through near-zero betas outside crises, Drobetz et al. and Feng and Zeng through their diversifier result, and Haffar and Le Fur through weak dependence with stock markets. None of them finds that CAT bonds are a hedge. The disagreement sits in one place, the behaviour of CAT bonds under financial stress. Carayannopoulos and Perez and Gürtler et al. find that CAT bonds moved with markets in the 2007-2009 crisis. Drobetz et al. find a strong safe haven against

stocks only after that crisis, Feng and Zeng find a safe haven after the COVID-19 outbreak, and Haffar and Le Fur find little extreme co-movement over their sample.

The negative findings come from the 2007-2009 crisis and the positive ones from later periods. Three explanations are available: the bonds changed, the crises differed, or the methods differed. My argument is that all three are true and that a simple decomposition can separate them.

3.2 A decomposition of the CAT bond return

I write the return of a CAT bond, or of an index of CAT bonds, over one period as

$$R_t = r_t + s - \ell_t - D \cdot \Delta\pi_t - k_t \tag{1}$$

where r_t is the return on the collateral, s is the contractual spread accrued over the period, ℓ_t is the loss caused by catastrophes (a realized loss, or a revaluation of the expected loss after an event), $\Delta\pi_t$ is the change in the premium investors require to hold the bond, D is the bond's spread duration, and k_t is the loss in collateral value caused by the failure of a counterparty or a fall in the value of the collateral assets.

The beta of the bond with respect to the market return R_m is the sum of the betas of its components, $\beta_x = \mathrm{Cov}(x_t, R_{m,t})/\mathrm{Var}(R_{m,t})$:

$$\beta = \beta_r - \beta_\ell - D \cdot \beta_\pi - \beta_k \tag{2}$$

The zero-beta argument concerns ℓ_t only: natural catastrophes are independent of financial markets, so $\beta_\ell = 0$. The collateral earns a short-term money-market rate whose weekly variation is negligible, so $\beta_r \approx 0$, and s is fixed by contract. What is left is

$$\beta \approx -D \cdot \beta_\pi - \beta_k \tag{3}$$

The measured beta of a natural-catastrophe bond is therefore the beta of its financial wrapper, not of its insurance risk. I call the first term the **premium channel** and the second the **collateral channel**. When markets fall, the required premium rises ($\beta_\pi < 0$) and collateral losses, if possible at all, become more likely ($\beta_k < 0$), so both channels push the beta above zero in a crisis.

Equation (3) leads to the following predictions.

1. In calm markets neither the premium nor the collateral moves with the market, so $\beta \approx 0$. This is the zero-beta result that every study finds outside crises.
2. The collateral channel is open only when the collateral is exposed to the credit of a financial institution. It was open before 2009 and is close to closed for deals whose collateral is held in Treasury instruments.
3. The premium channel opens when holders of CAT bonds need cash at the moment the market falls. They sell, and dealers short of capital absorb the bonds only at a higher premium. Brunnermeier and Pedersen (2009) show why this happens together with a market decline: when funding is scarce, market liquidity and funding liquidity reinforce each other. Herrmann and Hibbeln's (2023) estimate that liquidity accounts for about a

fifth of CAT bond spreads shows that this channel has room to move prices. The same channel covers the evidence of Gürtler et al. (2016) and Braun (2016): if the capital available to bear catastrophe risk shrinks with financial wealth, the price of catastrophe risk rises when credit spreads rise, as Froot's (2001) account of capital-driven catastrophe pricing would predict.

The decomposition also identifies the case in which the zero-beta claim fails at its root. If the peril is itself a macroeconomic shock, $\beta_\ell \neq 0$. For diversification purposes, pandemic bonds, and in my view bonds covering systemic cyber risk, belong to a different asset class from natural-catastrophe bonds.

3.3 Four stress regimes

Equation (3) suggests classifying stress episodes on two dimensions: whether the collateral was exposed to bank credit, and whether the episode was a funding crisis, in which investors needed cash, or a fall in equity prices while funding markets kept working. Table 2 places the evidence in this grid.

Table 2. Stress regimes implied by the decomposition, with the evidence for each

	Funding crisis (investors need cash)	**Equity sell-off, funding markets working**
Collateral exposed to bank credit (pre-2009 swap structures)	Both channels open: beta turns positive. *Evidence:* the subprime crisis after Lehman's failure (Carayannopoulos and Perez, 2015; Gürtler et al., 2016); the four Lehman-backed bonds.	Only the insurance core is tested: beta close to zero. *Evidence:* zero betas outside the crisis (Carayannopoulos and Perez, 2015).
Collateral in short-term Treasury instruments (post-2009)	Premium channel only: a small, short-lived drop; at most a weak safe haven. *Evidence:* March 2020 (a 0.81 percent fall in the week to 20 March, bid lists); safe haven against stocks after the outbreak (Feng and Zeng, 2025).	Neither channel open: beta close to zero, at least a weak safe haven. *Evidence:* strong post-crisis safe haven against stocks (Drobetz et al., 2020), possibly helped by index smoothing.

Each of the main results falls in one cell of the grid, so none of the studies needs to be wrong.

The top-left cell contains one episode, the months after Lehman's failure, when both channels were open. The crisis betas of Carayannopoulos and Perez and the link between CAT premiums and corporate spreads found by Gürtler et al. are what equation (3) predicts there. In that episode, much of the market exposure of CAT bonds came from a collateral design that the market has since abandoned.

The post-crisis safe haven of Drobetz et al. belongs to the bottom-right cell. After 2009 the collateral channel was closed, and in my reading the stock market declines of their post-crisis sample were not accompanied by forced selling on the scale of late 2008, so the premium channel stayed weak. The grid predicts a beta close to zero in this cell, which makes CAT bonds at least a weak safe haven. The strong form that Drobetz et al. report, a fall in the correlation during extreme weeks, needs more than that, and part of it may be statistical: an index of quotes that lags

the market shows a lower correlation in exactly the weeks when the market moves most (Section 3.4). In their crisis sub-period both channels were open, so the absence of a safe haven there is what the grid predicts. For the pre-crisis sub-period the grid predicts at most a weak safe haven, which is compatible with the absence of the strong form.

March 2020 is the test of the bottom-left cell: Treasury collateral combined with a severe scramble for cash. The framework predicts a small, short-lived fall driven by forced sales and no loss on collateral. That is what the market reports and Feng and Zeng's results describe. Prices fell by less than one percent in the week to 20 March, at the height of the scramble for cash, the markdowns came from bid lists of sellers, and Feng and Zeng still classify CAT bonds as a safe haven against stocks after the outbreak. The episode also shows why the statistical and economic meanings of a safe haven must be kept apart. A bond can qualify as a statistical safe haven in a regression on extreme weeks and still post a negative return in the week an investor needed to sell it.

Haffar and Le Fur's result reads, in this grid, as an average over the cells. An estimate over a full sample is dominated by calm weeks and, for post-2009 data, by the bottom row, so the dependence it measures is low. The grid predicts that the same copula estimated on the top-left cell alone would show stronger tail dependence.

3.4 Measurement choices that move the answer

The grid explains why the economic answer changes from one episode to the next. Part of the divergence between the articles is statistical, and here two biases pull in opposite directions.

The first is **smoothing**. If the index comes from bid indications that adjust with a lag, the observed return is a weighted average of current and past true returns, $R^{o}_{t} = \theta_0 R_t + \theta_1 R_{t-1} + \dots$ with weights summing to one (Getmansky et al., 2004). The contemporaneous covariance with the market is then θ_0 times the true covariance, so weekly betas and correlations are biased toward zero. The bias is probably largest in a crisis, when quotes are updated less often and θ_0 falls. In a regression on extreme weeks, the lag lowers the measured correlation in exactly the weeks that define a safe haven, so this bias makes CAT bonds look more like a safe haven than they are.

The second is **conditioning on volatility**. When correlations are estimated in a crisis window or in extreme weeks, the market's volatility is higher than usual, and since $\rho = \beta \cdot \sigma_m / \sigma_{CAT}$, the correlation rises even when the beta has not changed (Forbes and Rigobon, 2002). This bias makes CAT bonds look less like a safe haven than they are.

Which bias dominates depends on the method. A beta, as in Carayannopoulos and Perez, is free of the second bias but exposed to the first. A DCC regression on extreme weeks, as in Drobetz et al., is exposed to both. A copula on GARCH-filtered returns, as in Haffar and Le Fur, removes most of the second bias but still measures the dependence of observed, possibly smoothed, prices. Two further choices separate the studies: the comparison set (three bond and stock markets, six asset classes, or six stock indices) and the definition of stress (a calendar crisis window, extreme

weeks of the other asset, or the tails of a copula). With this many differences in design, the articles would be expected to disagree even if they studied the same market history.

3.5 Proposed solutions

The decomposition leads to two sets of proposals. The first concerns the design of future tests, the second the rules that can make the diversification benefit more reliable for investors.

A test design that addresses each source of divergence. Without estimating anything here, the framework specifies the test that would settle the disagreement rather than add one more estimate to it.

- *Separate the components of the return.* Swiss Re publishes a price return index alongside the total return index. The price return excludes coupons, so the collateral yield and the contractual spread drop out and what is left is mainly $-\ell_t - D \cdot \Delta\pi_t - k_t$. For post-2009 bonds, a stress beta should appear in this price component and nowhere else.
- *Measure funding stress directly.* Instead of a calendar crisis dummy or the quantiles of equity returns, the regression should include an indicator of funding conditions, such as the TED spread, next to the equity return. The prediction is that the safe-haven coefficient deteriorates when funding stress is high, holding the equity return fixed, and not when equities fall alone.
- *Correct both biases.* Desmooth the index, as Feng and Zeng (2025) do, adjust crisis correlations for volatility as Forbes and Rigobon (2002) propose, or use transaction prices from TRACE, as Herrmann and Hibbeln (2023) do.
- *Flag catastrophe weeks.* Weeks with a major catastrophe, such as the landfall of Hurricane Ike just before Lehman's failure, should be identified separately, so that a coincidence of the two shocks is not read as market beta.
- *Report results by regime and by liquidity.* Results should be reported cell by cell of Table 2. Since riskier bonds and bonds far from maturity are less liquid, the framework predicts larger stress betas for them.

Portfolio and market rules that remove what can be removed. Equation (3) also says which part of the beta can be engineered away.

- *Close the collateral channel.* Collateral should be held in short-term Treasury instruments in a trust account, with no bank guaranteeing its value and with the holdings disclosed. Most of the market has adopted this since 2009, but an investor should verify the collateral terms of each issue rather than assume them.
- *Contain the premium channel.* It cannot be eliminated, but investors can avoid feeding it. An investor that plans to raise cash by selling CAT bonds in a crisis creates the correlation it bought them to avoid. Funds that promise frequent redemptions should match those terms to the liquidity of their bonds, through notice periods, cash buffers or

limits on redemptions, so that they are not forced to sell into a falling market. At the portfolio level, the allocation should be small enough that it never has to be sold in a crisis.

- *Limit concentration within the allocation.* A portfolio concentrated in one peril and region, U.S. hurricane for instance, can lose several bonds to a single event, and Feng and Zeng (2025) find hedge ratios sensitive to extreme weather. Limits by peril, region and sponsor keep the insurance core diversified.
- *Classify perils honestly.* Pandemic bonds, and other bonds whose peril is itself a macroeconomic shock, should not be presented or modelled as zero-beta exposures.

3.6 Conclusion of the analysis

Whether CAT bonds are zero-beta assets depends on which part of the bond the question refers to. For natural perils, the insurance core is zero-beta. The bond as a traded security has a beta close to zero in calm markets that can turn positive in a funding crisis through the premium channel, and that was turned positive in 2008 by a collateral design the market has since abandoned. On the evidence reviewed, CAT bonds are a reliable diversifier, they are not a hedge, and their safe-haven property is conditional: present in equity sell-offs, weaker in funding crises, and absent for perils that are themselves macroeconomic shocks.

4. Conclusion

This paper asked whether CAT bonds keep their diversification benefits during financial stress. Carayannopoulos and Perez (2015) find that they lost their zero-beta property after Lehman's failure. Drobetz et al. (2020) find that they are a diversifier in every period but a strong safe haven only after the financial crisis, and Haffar and Le Fur (2022) find little extreme co-movement with equity markets. Gürtler et al. (2016) and Braun (2016) add evidence that CAT bond premiums move with corporate spreads, while Feng and Zeng (2025) find a safe haven after the COVID-19 outbreak.

My contribution is to show that these conclusions are compatible once the CAT bond return is split into an insurance core and a financial wrapper. The core is independent of markets for natural perils. The wrapper is not, and its exposure depends on the collateral structure and on whether a stress episode is a funding crisis. Sorting the evidence into the four regimes of Table 2 explains why the 2007-2009 crisis produced positive betas, why the post-crisis period produced a safe haven against stocks, and why March 2020 produced a small, temporary fall. A further part of the disagreement is statistical: quote-based indices bias co-movement toward zero, crisis-window correlations bias it upward, and the studies are exposed to these biases in different degrees.

The answer to the research question is therefore conditional. CAT bonds retain their diversification benefit in financial stress when their collateral is safe and their holders are not forced to sell. Under those conditions they can act as a safe haven against stocks. They lose part

of it when either condition fails. The practical implication follows: investors can make the benefit more reliable through collateral rules, liquidity management and concentration limits, but they should treat CAT bonds as a diversifier with a conditional safe-haven property, not as crisis insurance.

The analysis has limits. It rests on published results rather than new estimates, so the four regimes and the predictions of Section 3.5 are hypotheses that still need to be tested. The post-2009 period contains only one severe funding crisis, March 2020, so the evidence for the bottom-left cell of Table 2 is thin. Finally, the framework assumes that catastrophe losses are independent of markets for natural perils. That assumption is the right starting point today, but it deserves to be re-examined as pandemic, cyber and climate-related risks take a larger share of the market.

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